

MASVINGO, Zimbabwe

WMO: 679750

Lat: 20.055S

Lon: 30.859E

Elev: 3595

StdP: 12.88

Time Zone: 2.00 (E02)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.2	41.5	24.3	21.1	68.5	27.7	24.7	67.7	21.0	61.4	19.6	61.8	1.8	350	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	92.7	63.7	90.8	63.7	88.7	63.4	70.8	80.0	70.0	79.0	69.2	78.0	9.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	119.6	72.6	67.8	116.8	72.1	67.1	114.0	71.8	37.1	80.1	36.4	79.5	35.7	77.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	18.2	16.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	73.3	72.3	71.1	67.0	61.8	57.9	57.4	62.3	67.9	71.6	73.2	72.5
	DBStd	7.30	3.54	3.31	3.86	4.22	4.18	4.42	3.80	5.20	5.29	5.92	4.95	4.21
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	774	1	0	4	26	117	216	235	119	31	16	5	4
	CDD50	6327	722	626	654	510	366	238	232	380	536	669	695	699
	CDD65	1625	258	206	193	86	18	3	1	34	118	220	250	238
	CDH74	15522	1924	1624	1486	790	411	163	120	652	1551	2480	2387	1934
	CDH80	5281	639	511	440	201	81	12	5	154	549	1087	960	642
(14) Wind	WSAvg	6.6	6.8	6.2	6.6	6.0	5.3	5.6	6.0	6.7	8.0	8.2	7.2	6.5
(15) Precipitation	PrecAvg	23.8	5.0	4.1	2.4	1.0	0.4	0.2	0.1	0.2	0.4	1.1	3.1	5.7
	PrecMax	35.7	17.2	11.8	10.5	5.0	2.2	2.0	1.2	0.9	2.2	3.0	9.0	15.9
	PrecMin	8.8	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.9
	PrecStd	7.2	4.3	3.1	2.5	1.1	0.6	0.4	0.2	0.2	0.6	0.9	2.0	4.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	93.4	91.7	89.7	87.3	81.9	80.6	88.4	92.8	95.5	95.0	92.6
		MCWB	65.6	65.6	65.7	64.1	60.5	58.5	54.9	59.4	61.6	63.9	64.0	64.9
	2%	DB	89.1	89.6	88.0	85.5	82.6	78.5	77.5	84.9	89.5	92.1	91.8	89.5
		MCWB	65.9	66.1	65.5	64.2	59.2	56.6	54.7	58.5	59.9	62.3	63.4	65.9
	5%	DB	86.3	86.2	85.6	81.8	79.1	75.5	74.8	81.5	86.8	89.5	89.0	86.5
		MCWB	66.5	66.1	65.6	63.6	58.0	55.3	54.3	57.3	59.0	61.8	63.5	65.5
	10%	DB	83.7	83.6	82.8	78.6	75.8	72.4	71.9	78.1	83.9	86.3	85.7	83.8
		MCWB	67.0	66.4	66.1	63.0	57.4	54.5	53.7	56.1	58.5	61.6	63.9	65.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	71.4	71.4	68.9	63.2	61.0	59.1	63.0	65.8	68.2	70.2	71.0
		MCDB	81.2	80.5	81.0	79.5	76.4	75.2	70.6	80.4	79.8	82.1	83.1	80.7
	2%	WB	70.8	70.1	69.9	67.4	61.7	59.2	57.3	60.8	63.8	66.6	68.8	70.0
		MCDB	79.2	79.4	79.0	77.6	75.6	73.2	70.3	77.7	79.1	81.0	80.0	79.4
	5%	WB	69.9	69.3	68.8	66.2	60.3	57.5	56.2	59.2	62.3	65.5	67.5	69.0
		MCDB	78.0	78.2	77.4	75.4	73.9	70.8	69.3	75.9	78.2	80.1	78.0	78.2
	10%	WB	69.1	68.5	67.8	65.0	59.1	56.0	54.9	57.8	60.8	64.4	66.6	68.1
		MCDB	76.6	76.8	76.1	73.4	71.9	68.4	67.9	74.1	77.2	79.0	76.6	76.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.8	19.0	21.2	26.3	27.3	26.3	27.6	26.6	23.7	20.6	19.0
		MCDBR	24.2	25.2	25.5	27.4	32.0	32.9	33.6	32.5	31.9	30.7	27.1	24.2
		MCWBR	7.1	7.4	8.0	9.4	12.3	13.9	14.5	11.5	10.7	9.1	7.7	6.6
	5% WB	MCDBR	15.9	16.6	18.0	20.4	25.5	27.2	25.8	27.3	24.9	23.3	17.7	18.2
		MCWBR	5.6	5.8	7.0	8.1	11.3	12.2	12.1	10.4	8.9	7.8	6.3	6.2
(40) Clear-Sky Solar Irradiance	taub		0.353	0.349	0.338	0.310	0.285	0.281	0.289	0.326	0.400	0.400	0.379	0.366
	taud		2.495	2.510	2.542	2.585	2.606	2.592	2.546	2.389	2.208	2.244	2.359	2.450
	Ebn at Noon		314	312	308	305	301	297	297	294	284	292	304	310
	Edh at Noon		37	36	33	30	27	26	28	36	46	46	42	39
	RadAvg		2038	2001	1793	1643	1525	1341	1367	1692	1899	2040	1982	1940
(45) All-Sky Solar Radiation	RadStd		159	193	158	102	89	101	99	62	102	124	144	197

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page